

Network Configuration Compliance Assessment

0017C573E288 — sonicwall NSA 3700

1. Device identification

Hostname	0017C573E288
Vendor	sonicwall
Model	NSA 3700
Operating system	SonicOS
Software version	7.3.0-7012-R8150
Serial number	18C241359276
Platform identifier	sonicwall_sonicos

Source of assessment

Configuration file	sonicwall config file.txt
SHA-256 of file	b6be6eb3e12089f17b62f9edb7f27775a1bffe9c16e9ead9760380f8c91aba4f
Records in file	92,635
Assessment identifier	NCSA-2026-0001
Report generated	10 September 2026 at 08:55 UTC

The SHA-256 digest identifies the exact file assessed. A re-run against a different file will produce a different digest, and the two reports are not comparable.

2. Result

Compliance score	36.2% of 47 decided controls
Assessment coverage	53.4% of 88 controls in the catalogue
Controls not decided	20
Controls not applicable	21
Aggregate risk	154.8 — highest band CRITICAL

How to read the score. The compliance score is calculated over controls that could be decided, and it is reported beside coverage for that reason. A high score on low coverage means little of the device could be assessed — not that the device is well configured. Both figures must be quoted together.

3. Result states

Every control resolves to exactly one of seven states. The distinction between *not applicable* and *unknown* is material: the first is a property of the device, the second is a limit of this assessment.

State	Controls	Share	Meaning
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PASS	17	19.3%	Checked; the device is configured correctly.
FAIL	28	31.8%	Checked; the device is not configured correctly.
PARTIAL	2	2.3%	Checked; some scoped instances comply and others do not.
NOT_APPLICABLE	21	23.9%	The control does not apply to this device. Excluded from scoring; a justification is recorded for each.
UNKNOWN	20	22.7%	Could not be determined from the supplied configuration. Neither a pass nor a failure.
MANUAL_REVIEW	0	0.0%	Requires a human decision; no automated verdict would be sound.
ERROR	0	0.0%	The check itself failed. A defect in the auditing tool, not in the device.

4. Findings

30 controls assessed as failing or partially met, ordered by severity. Each is reported with the configuration line it was derived from.

Severity	Findings
Critical	2
High	12
Medium	13
Low	3

4.1 Critical severity (2)

NCSA-CAT-002 — Firewall rules must not permit any source to any destination

FAIL **CRITICAL** firewall.rules.source

Observed 0.0.0.0/0

Required 0.0.0.0/0, any

Risk 15.0 (CRITICAL) — severity critical -> base 10.0; exposure x1.5: reachable from an untrusted zone in the policy; confidence x1.0: value observed in the configuration

Framework references NIST SP 800-53: AC-4, SC-7.5; ISO/IEC 27001: A.5.14, A.8.16, A.8.20, A.8.22, A.8.23

Basis. worst of 329 scoped instance(s); FAIL at Test_SSH [IPv4#1, Custom Access Rule [IPv4#2, Malicious_IP [IPv4#3

Evidence

Setting	Configuration line
37290	policyName_1=Test_SSH

Source: sonicwall config file.txt

This configuration is one physical line of &-separated settings, so every value sits on line 1 and a line number would distinguish nothing. The figure above is the setting's position in the file. To read setting *n* directly: `tr '&' '\n' < config | sed -n 'np'`

NCSA-CLD-001 — Administrative ports must not be open to the internet

FAIL **CRITICAL** exposure.admin_ports_open_to_internet

Observed

rule Default Access Rule [IPv4#68]: http-mgmt/80 from internet to 45.64.204.198, rule Default Access Rule [IPv4#71]: http-mgmt/80 from internet to 45.64.204.194, rule Default Access Rule [IPv4#72]: http-mgmt/80 from internet to 123.252.251.38, rule Default Access Rule [IPv4#73]: http-mgmt/80 from internet to 182.78.55.110, rule Default Access Rule [IPv4#80]: http-mgmt/80 from internet to 115.242.234.130, rule Default Access Rule [IPv4#82]: http-mgmt/80 from internet to 125.99.39.178 ... and 1 more

Required

0

Risk

15.0 (CRITICAL) — severity critical -> base 10.0; exposure x1.5: reachable from an untrusted zone in the policy; confidence x1.0: value observed in the configuration

Framework references

NIST SP 800-53: AC-17(1), SC-7, SC-7(5); ISO/IEC 27001: A.5.14, A.6.7, A.8.16, A.8.20, A.8.22, A.8.23

Evidence

Setting	Configuration line
42784	policyName_68=Default Access Rule
43030	policyName_71=Default Access Rule
43112	policyName_72=Default Access Rule
43194	policyName_73=Default Access Rule
43768	policyName_80=Default Access Rule
43932	policyName_82=Default Access Rule
44178	policyName_85=Default Access Rule

Source:sonicwall config file.txt

This configuration is one physical line of &-separated settings, so every value sits on line 1 and a line number would distinguish nothing. The figure above is the setting's position in the file. To read setting *n* directly: `tr '&' '\n' < config | sed -n 'np'`

4.2 High severity (12)

NCSA-CAT-004 — All policy references must resolve

FAIL **HIGH** firewall.rules_unevaluable

Observed	Default Access Rule [IPv4#0]: SonicWALL SSO Agents [UNRESOLVED]; SonicWALL SSO Agents [UNRESOLVED], Default Access Rule [IPv4#11]: Ping [UNSUPPORTED], Default Access Rule [IPv4#14]: Ping [UNSUPPORTED], Default Access Rule [IPv4#17]: Ping [UNSUPPORTED], Default Access Rule [IPv4#21]: Ping [UNSUPPORTED], Default Access Rule [IPv4#46]: WAN RemoteAccess Networks [UNRESOLVED] ... and 43 more
Required	0
Risk	9.0 (HIGH) — severity high -> base 6.0; exposure x1.5: reachable from an untrusted zone in the policy; confidence x1.0: value observed in the configuration
Framework references	NIST SP 800-53: CM-6, AC-4; ISO/IEC 27001: A.5.14, A.8.22, A.8.23, A.8.9

Evidence

Setting	Configuration line
37208	rule Default Access Rule [IPv4#0] not evaluable: SonicWALL SSO Agents [UNRESOLVED]; SonicWALL SSO Agents [UNRESOLVED]

Source: sonicwall config file.txt

This configuration is one physical line of &-separated settings, so every value sits on line 1 and a line number would distinguish nothing. The figure above is the setting's position in the file. To read setting *n* directly: `tr '&' '\n' < config | sed -n 'np'`

NCSA-EXT-001 — SSH must use strong key exchange and ciphers

FAIL **HIGH** crypto.weak_ciphers

Observed	sw_kexalgs:diffie-hellman-group1-sha1, sw_kexalgs:diffie-hellman-group14-sha1, sw_kexalgs:diffie-hellman-group-exchange-sha1, sw_macs: hmac-sha1
Required	0
Risk	6.0 (MEDIUM) — severity high -> base 6.0; exposure x1.0: no untrusted-zone path proven for this control; confidence x1.0: value observed in the configuration
Framework references	NIST SP 800-53: SC-8.1, SC-13; ISO/IEC 27001: A.5.10, A.5.14, A.5.33, A.8.20, A.8.24, A.8.26

Evidence

Setting	Configuration line
35468	sshCipherControlConfig={"sw_kexalgs": [{"diffie-hellman-group1-sha1": 1}, {"diffie-hellman-group14-sha1": 1}, {"diffie-hellman-group-exchange-sha1": 1}, {"diffie-hellman-group-exchange-sha256": 1}, {"

Source: sonicwall config file.txt

This configuration is one physical line of &-separated settings, so every value sits on line 1 and a line number would distinguish nothing. The figure above is the setting's position in the file. To read setting *n* directly: `tr '&' '\n' < config | sed -n 'np'`

NCSA-EXT-013 — Account lockout must be enabled

FAIL **HIGH** authentication.lockout.enabled

Observed	false
Required	true
Risk	6.0 (MEDIUM) — severity high -> base 6.0; exposure x1.0: no untrusted-zone path proven for this control; confidence x1.0: value observed in the configuration
Framework references	NIST SP 800-53: AC-7; ISO/IEC 27001: A.8.5

Evidence

Setting	Configuration line
21225	loginFailLockoutEnable=off

Source: sonicwall config file.txt

This configuration is one physical line of &-separated settings, so every value sits on line 1 and a line number would distinguish nothing. The figure above is the setting's position in the file. To read setting *n* directly: `tr '&' '\n' < config | sed -n 'np'`

NCSA-EXT-014 — Administrative MFA should be required

FAIL **HIGH** authentication.mfa.admin_required

Observed false
Required true
Risk 6.0 (MEDIUM) — severity high -> base 6.0; exposure x1.0: no untrusted-zone path proven for this control; confidence x1.0: value observed in the configuration
Framework references NIST SP 800-53: IA-2.1; ISO/IEC 27001: A.5.16

Evidence

Setting	Configuration line
21234	adminLoginOtpRequire=0

Source: sonicwall config file.txt

This configuration is one physical line of &-separated settings, so every value sits on line 1 and a line number would distinguish nothing. The figure above is the setting's position in the file. To read setting *n* directly: `tr '&' '\n' < config | sed -n 'np'`

NCSA-EXT-022 — Remote syslog must be configured

FAIL **HIGH** logging.remote_syslog

Observed false
Required true
Risk 6.0 (MEDIUM) — severity high -> base 6.0; exposure x1.0: no untrusted-zone path proven for this control; confidence x1.0: value observed in the configuration
Framework references NIST SP 800-53: AU-4, AU-9; ISO/IEC 27001: A.5.33, A.8.15, A.8.6

Evidence

Setting	Configuration line
21291	syslogServerName=

Source: sonicwall config file.txt

This configuration is one physical line of &-separated settings, so every value sits on line 1 and a line number would distinguish nothing. The figure above is the setting's position in the file. To read setting *n* directly: `tr '&' '\n' < config | sed -n 'np'`

NCSA-EXT-023 — SNMPv3 authentication must be enabled

FAIL **HIGH** snmp.v3_auth

Observed	false
Required	true
Risk	9.0 (HIGH) — severity high -> base 6.0; exposure x1.5: management plane sits behind an untrusted-zone rule; confidence x1.0: value observed in the configuration
Framework references	NIST SP 800-53: IA-3, SC-8; ISO/IEC 27001: A.5.10, A.5.14, A.5.33, A.8.20, A.8.26

Evidence

Setting	Configuration line
23686	Snmp3_Mand_Required=off

Source: sonicwall config file.txt

This configuration is one physical line of &-separated settings, so every value sits on line 1 and a line number would distinguish nothing. The figure above is the setting's position in the file. To read setting *n* directly: `tr '&' '\n' < config | sed -n 'np'`

NCSA-EXT-025 — SNMP access must be restricted by ACL

FAIL **HIGH** snmp.community_acl

Observed	—
Required	true
Risk	6.3 (MEDIUM) — severity high -> base 6.0; exposure x1.5: management plane sits behind an untrusted-zone rule; confidence x0.7: no evidence anchor; value not directly observed
Framework references	NIST SP 800-53: AC-3, SC-7; ISO/IEC 27001: A.5.14, A.5.15, A.5.33, A.8.16, A.8.18, A.8.20, A.8.22, A.8.23, A.8.26, A.8.3, A.8.4

Basis. worst of 36 scoped instance(s); FAIL at snmp_mgmt_src_addr_0, snmp_mgmt_src_addr_1, snmp_mgmt_src_addr_2

Evidence

No line in the configuration carries this setting. The finding rests on its absence, which the basis above states. There is no position to cite, because there is nothing there.

NCSA-EXT-039 — Intrusion prevention should be enabled

FAIL **HIGH** security.ips.enabled

Observed —
Required true
Risk 4.199999999999999 (MEDIUM) — severity high -> base 6.0; exposure x1.0: no untrusted-zone path proven for this control; confidence x0.7: no evidence anchor; value not directly observed
Framework references NIST SP 800-53: SI-4; ISO/IEC 27001: A.8.16

Basis. security.ips.enabled not present in configuration

Evidence

No line in the configuration carries this setting. The finding rests on its absence, which the basis above states. There is no position to cite, because there is nothing there.

NCSA-PLT-002 — Stored credentials and configuration parameters must be encrypted

FAIL **HIGH** platform.stored_credentials_encrypted

Observed false
Required true
Risk 6.0 (MEDIUM) — severity high -> base 6.0; exposure x1.0: no untrusted-zone path proven for this control; confidence x1.0: value observed in the configuration
Framework references NIST SP 800-53: IA-5(1), SC-28; ISO/IEC 27001: A.5.17, A.8.24

Evidence

Setting	Configuration line
181	encUsernamePassword=off

Source: sonicwall config file.txt

This configuration is one physical line of &-separated settings, so every value sits on line 1 and a line number would distinguish nothing. The figure above is the setting's position in the file. To read setting *n* directly: `tr '&' '\n' < config | sed -n 'np'`

NCSA-PWD-001 — Minimum password length must be enforced

FAIL **HIGH** authentication.min_password_length

Observed 1

Required 15

Risk 6.0 (MEDIUM) — severity high -> base 6.0; exposure x1.0: no untrusted-zone path proven for this control; confidence x1.0: value observed in the configuration

Framework references NIST SP 800-53: IA-5, IA-5(1); ISO/IEC 27001: A.5.16, A.5.17

Evidence

Setting	Configuration line
21208	minPasswordLength=1

Source: sonicwall config file.txt

This configuration is one physical line of &-separated settings, so every value sits on line 1 and a line number would distinguish nothing. The figure above is the setting's position in the file. To read setting *n* directly: `tr '&' '\n' < config | sed -n 'np'`

NCSA-SNMP-001 — SNMP v1/v2c must not be used

FAIL **HIGH** snmp.version

Observed 2c

Required 1, v1, 2c, v2c, 2

Risk 9.0 (HIGH) — severity high -> base 6.0; exposure x1.5: management plane sits behind an untrusted-zone rule; confidence x1.0: value observed in the configuration

Framework references NIST SP 800-53: AC-17(2), IA-3, SC-8; ISO/IEC 27001: A.5.10, A.5.14, A.5.33, A.6.7, A.8.20, A.8.26

Evidence

Setting	Configuration line
23676	snmp_GetCommunity=HUB

Source: sonicwall config file.txt

This configuration is one physical line of &-separated settings, so every value sits on line 1 and a line number would distinguish nothing. The figure above is the setting's position in the file. To read setting *n* directly: `tr '&' '\n' < config | sed -n 'np'`

NCSA-TIME-001 — Idle session timeout must be 10 minutes or less

FAIL **HIGH** management.vty.exec_timeout

Observed 120

Required 10

Risk 9.0 (HIGH) — severity high -> base 6.0; exposure x1.5: management plane sits behind an untrusted-zone rule; confidence x1.0: value observed in the configuration

Framework references NIST SP 800-53: AC-12, SC-10; ISO/IEC 27001: A.8.20

Evidence

Setting	Configuration line
21204	adminLoginTimeout=120

Source: sonicwall config file.txt

This configuration is one physical line of &-separated settings, so every value sits on line 1 and a line number would distinguish nothing. The figure above is the setting's position in the file. To read setting *n* directly: `tr '&' '\n' < config | sed -n 'np'`

4.3 Medium severity (13)

NCSA-BAN-001 — A login banner must be displayed

FAIL **MEDIUM** banner.login

Observed —

Required —

Risk 3.15 (MEDIUM) — severity medium -> base 3.0; exposure x1.5: management plane sits behind an untrusted-zone rule; confidence x0.7: no evidence anchor; value not directly observed

Framework references NIST SP 800-53: AC-8; ISO/IEC 27001: A.8.5

Basis. banner.login not present in configuration

Evidence

No line in the configuration carries this setting. The finding rests on its absence, which the basis above states. There is no position to cite, because there is nothing there.

NCSA-CAT-006 — Access control lists must be defined

FAIL **MEDIUM** firewall.acls

Observed	—
Required	—
Risk	3.15 (MEDIUM) — severity medium -> base 3.0; exposure x1.5: reachable from an untrusted zone in the policy; confidence x0.7: no evidence anchor; value not directly observed
Framework references	NIST SP 800-53: AC-4; ISO/IEC 27001: A.5.14, A.8.22, A.8.23

Basis. firewall.acls not present in configuration

Evidence

No line in the configuration carries this setting. The finding rests on its absence, which the basis above states. There is no position to cite, because there is nothing there.

NCSA-CAT-010 — Interfaces must be assigned to a zone

FAIL **MEDIUM** interfaces.zone

Observed	—
Required	—
Risk	2.0999999999999996 (LOW) — severity medium -> base 3.0; exposure x1.0: no untrusted-zone path proven for this control; confidence x0.7: no evidence anchor; value not directly observed
Framework references	NIST SP 800-53: SC-7; ISO/IEC 27001: A.5.14, A.8.16, A.8.20, A.8.22, A.8.23

Basis. worst of 36 scoped instance(s); FAIL at interface_Zone_6, interface_Zone_9, interface_Zone_13

Evidence

No line in the configuration carries this setting. The finding rests on its absence, which the basis above states. There is no position to cite, because there is nothing there.

NCSA-EXT-006 — SSH idle sessions must time out

FAIL **MEDIUM** `management.ssh.timeout`

Observed 300

Required 120

Risk 4.5 (MEDIUM) — severity medium -> base 3.0; exposure x1.5: management plane sits behind an untrusted-zone rule; confidence x1.0: value observed in the configuration

Framework references NIST SP 800-53: AC-12, SC-10; ISO/IEC 27001: A.8.20

Evidence

Setting	Configuration line
182	<code>cli_idleTimeout=300</code>

Source: `sonicwall config file.txt`

This configuration is one physical line of &-separated settings, so every value sits on line 1 and a line number would distinguish nothing. The figure above is the setting's position in the file. To read setting *n* directly: `tr '&' '\n' < config | sed -n 'np'`

NCSA-EXT-007 — HTTPS management must be enabled where web management is used

FAIL **MEDIUM** `management.https.enabled`

Observed false

Required true

Risk 4.5 (MEDIUM) — severity medium -> base 3.0; exposure x1.5: management plane sits behind an untrusted-zone rule; confidence x1.0: value observed in the configuration

Framework references NIST SP 800-53: SC-8.1; ISO/IEC 27001: A.5.10, A.5.14, A.5.33, A.8.20, A.8.26

Evidence

Setting	Configuration line
21029	<code>wseIsHttpsEnabled=off</code>

Source: `sonicwall config file.txt`

This configuration is one physical line of &-separated settings, so every value sits on line 1 and a line number would distinguish nothing. The figure above is the setting's position in the file. To read setting *n* directly: `tr '&' '\n' < config | sed -n 'np'`

NCSA-EXT-008 — HTTP management must redirect to HTTPS

FAIL **MEDIUM** management.http.redirect_https

Observed	false
Required	true
Risk	4.5 (MEDIUM) — severity medium -> base 3.0; exposure x1.5: management plane sits behind an untrusted-zone rule; confidence x1.0: value observed in the configuration
Framework references	NIST SP 800-53: SC-8.1; ISO/IEC 27001: A.5.10, A.5.14, A.5.33, A.8.20, A.8.26

Basis. worst of 36 scoped instance(s); FAIL at iface_http_redirect_rule_6

Evidence

Setting	Configuration line
65374	iface_http_redirect_rule_6=0

Source: sonicwall config file.txt

This configuration is one physical line of &-separated settings, so every value sits on line 1 and a line number would distinguish nothing. The figure above is the setting's position in the file. To read setting *n* directly: `tr '&' '\n' < config | sed -n 'np'`

NCSA-EXT-021 — Denied traffic must be logged

FAIL **MEDIUM** logging.log_denied

Observed	false
Required	true
Risk	3.0 (MEDIUM) — severity medium -> base 3.0; exposure x1.0: no untrusted-zone path proven for this control; confidence x1.0: value observed in the configuration
Framework references	NIST SP 800-53: AU-2, AC-4; ISO/IEC 27001: A.5.14, A.8.15, A.8.22, A.8.23

Evidence

Setting	Configuration line
21231	loginFailLockoutLog=off

Source: sonicwall config file.txt

This configuration is one physical line of &-separated settings, so every value sits on line 1 and a line number would distinguish nothing. The figure above is the setting's position in the file. To read setting *n* directly: `tr '&' '\n' < config | sed -n 'np'`

NCSA-EXT-029 — Configuration change archiving should be enabled

FAIL **MEDIUM** platform.config_archive

Observed false
Required true
Risk 3.0 (MEDIUM) — severity medium -> base 3.0; exposure x1.0: no untrusted-zone path proven for this control; confidence x1.0: value observed in the configuration
Framework references NIST SP 800-53: CM-3, CM-6; ISO/IEC 27001: 8.1, 9.3.3, A.8.32, A.8.9

Evidence

Setting	Configuration line
35452	backupStorage=0

Source: sonicwall config file.txt

This configuration is one physical line of &-separated settings, so every value sits on line 1 and a line number would distinguish nothing. The figure above is the setting's position in the file. To read setting *n* directly: `tr '&' '\n' < config | sed -n 'np'`

NCSA-EXT-040 — Gateway anti-virus should be enabled

FAIL **MEDIUM** security.gav.enabled

Observed —
Required true
Risk 2.0999999999999996 (LOW) — severity medium -> base 3.0; exposure x1.0: no untrusted-zone path proven for this control; confidence x0.7: no evidence anchor; value not directly observed
Framework references NIST SP 800-53: SI-3; ISO/IEC 27001: A.8.7

Basis. security.gav.enabled not present in configuration

Evidence

No line in the configuration carries this setting. The finding rests on its absence, which the basis above states. There is no position to cite, because there is nothing there.

NCSA-LOG-002 — At least two remote syslog servers must be configured

PARTIAL **MEDIUM** logging.servers

Observed false

Required 2

Framework references NIST SP 800-53: AU-4, AU-9, AU-6; ISO/IEC 27001: A.5.25, A.5.33, A.6.8, A.8.15, A.8.6

Basis. partially satisfied: observed False, expected 2

Evidence

Setting	Configuration line
21291	syslogServerName=

Source: sonicwall config file.txt

This configuration is one physical line of &-separated settings, so every value sits on line 1 and a line number would distinguish nothing. The figure above is the setting's position in the file. To read setting *n* directly: `tr '&' '\n' < config | sed -n 'np'`

NCSA-PWD-002 — Password complexity must be enforced

FAIL **MEDIUM** authentication.complexity

Observed false

Required true

Risk 3.0 (MEDIUM) — severity medium -> base 3.0; exposure x1.0: no untrusted-zone path proven for this control; confidence x1.0: value observed in the configuration

Framework references NIST SP 800-53: IA-5(1); ISO/IEC 27001: A.5.16, A.5.17

Evidence

Setting	Configuration line
21220	passwordComplexity=0

Source: sonicwall config file.txt

This configuration is one physical line of &-separated settings, so every value sits on line 1 and a line number would distinguish nothing. The figure above is the setting's position in the file. To read setting *n* directly: `tr '&' '\n' < config | sed -n 'np'`

NCSA-SSH-003 — Weak SSH ciphers must not be permitted

FAIL **MEDIUM** crypto.weak_ciphers

Observed	sw_kexalgs:diffie-hellman-group1-sha1, sw_kexalgs:diffie-hellman-group14-sha1, sw_kexalgs:diffie-hellman-group-exchange-sha1, sw_macs:hmac-sha1
Required	0
Risk	3.0 (MEDIUM) — severity medium -> base 3.0; exposure x1.0: no untrusted-zone path proven for this control; confidence x1.0: value observed in the configuration
Framework references	NIST SP 800-53: SC-8(1), SC-13; ISO/IEC 27001: A.5.10, A.5.14, A.5.33, A.8.20, A.8.24, A.8.26

Evidence

Setting	Configuration line
35468	sshCipherControlConfig={"sw_kexalgs": [{"diffie-hellman-group1-sha1": 1}, {"diffie-hellman-group14-sha1": 1}, {"diffie-hellman-group-exchange-sha1": 1}, {"diffie-hellman-group-exchange-sha256": 1}, {"

Source: sonicwall config file.txt

This configuration is one physical line of &-separated settings, so every value sits on line 1 and a line number would distinguish nothing. The figure above is the setting's position in the file. To read setting *n* directly: `tr '&' '\n' < config | sed -n 'np'`

NCSA-SSH-004 — SSH authentication attempts must be limited

FAIL **MEDIUM** management.ssh.max_auth_tries

Observed	5
Required	3
Risk	4.5 (MEDIUM) — severity medium -> base 3.0; exposure x1.5: management plane sits behind an untrusted-zone rule; confidence x1.0: value observed in the configuration
Framework references	NIST SP 800-53: AC-7; ISO/IEC 27001: A.8.5

Evidence

Setting	Configuration line
21210	loginAttemptCLI=5

Source: sonicwall config file.txt

This configuration is one physical line of &-separated settings, so every value sits on line 1 and a line number would distinguish nothing. The figure above is the setting's position in the file. To read setting *n* directly: `tr '&' '\n' < config | sed -n 'np'`

4.4 Low severity (3)

NCSA-EXT-027 — At least two time sources should be configured

PARTIAL **LOW** time.servers

Observed

Required 2

Framework references NIST SP 800-53: AU-8; ISO/IEC 27001: A.8.17

Basis. partially satisfied: observed [""], expected 2

Evidence

Setting	Configuration line
21188	ntp_srv_intfs=

Source: sonicwall config file.txt

This configuration is one physical line of &-separated settings, so every value sits on line 1 and a line number would distinguish nothing. The figure above is the setting's position in the file. To read setting *n* directly: `tr '&' '\n' < config | sed -n 'np'`

NCSA-EXT-030 — A MOTD banner should be present

FAIL **LOW** banner.motd

Observed —

Required —

Risk 1.049999999999998 (LOW) — severity low -> base 1.0; exposure x1.5: management plane sits behind an untrusted-zone rule; confidence x0.7: no evidence anchor; value not directly observed

Framework references NIST SP 800-53: AC-8; ISO/IEC 27001: A.8.5

Basis. banner.motd not present in configuration

Evidence

No line in the configuration carries this setting. The finding rests on its absence, which the basis above states. There is no position to cite, because there is nothing there.

NCSA-EXT-042 — Interfaces should carry a description

FAIL **LOW** interfaces.description

Observed	—
Required	—
Risk	0.7 (LOW) — severity low -> base 1.0; exposure x1.0: no untrusted-zone path proven for this control; confidence x0.7: no evidence anchor; value not directly observed
Framework references	NIST SP 800-53: CM-8; ISO/IEC 27001: A.5.9, A.8.9

Basis. worst of 36 scoped instance(s); FAIL at iface_comment_2, iface_comment_4, iface_comment_6

Evidence

No line in the configuration carries this setting. The finding rests on its absence, which the basis above states. There is no position to cite, because there is nothing there.

5. Controls not applicable (21)

These controls are excluded from the compliance score. Because an exclusion raises the score by removing a control from the denominator, each one carries a stated justification. An exclusion without a justification is a defect and is treated as such.

Control	Title	Justification
NCSA-API-001	Management API must not accept HTTP basic authentication	the management REST API is disabled, so its authentication settings are moot
NCSA-API-002	Management API digest authentication must not use MD5	the management REST API is disabled, so its authentication settings are moot
NCSA-API-003	Management API public keys must be at least 2048 bits	the management REST API is disabled, so its authentication settings are moot
NCSA-API-004	Management API must not permit unrestricted cross-origin requests	the management REST API is disabled, so its authentication settings are moot
NCSA-API-005	Management API session security must be enabled	the management REST API is disabled, so its authentication settings are moot
NCSA-API-006	Management API access must be logged	the management REST API is disabled, so its authentication settings are moot
NCSA-CAT-008	Cloud security groups must be identified	platform 'sonicwall_sonicos' has no 'cloud' capability: this is an on-premises appliance, not a cloud resource
NCSA-CAT-009	Egress traffic should be restricted	platform 'sonicwall_sonicos' has no 'cloud' capability: this is an on-premises appliance, not a cloud resource
NCSA-EXT-002	SSH strict host key checking should be enabled	management.ssh.strict_host_key does not exist on platform 'sonicwall_sonicos': SonicOS is not acting as an SSH client here
NCSA-EXT-004	Public-key authentication should be available for SSH	management.ssh.pubkey_auth does not exist on platform 'sonicwall_sonicos': admin SSH authenticates by password or TOTP, not public key
NCSA-EXT-005	RSA host key must be at least 2048 bits	crypto.rsa_modulus does not exist on platform 'sonicwall_sonicos': key size is not represented anywhere in the export
NCSA-EXT-009	The auxiliary port must not offer an EXEC session	management.aux.exec_disabled does not exist on platform 'sonicwall_sonicos': the appliance has no AUX port
NCSA-EXT-010	Management lines must not permit outbound sessions	management.vty.transport_output does not exist on platform 'sonicwall_sonicos': no VTY lines; management is zone-bound
NCSA-EXT-012	A privileged-mode secret must be set, not a reversible password	authentication.enable_secret does not exist on platform 'sonicwall_sonicos': no IOS-style enable secret on SonicOS
NCSA-EXT-034	DHCP snooping should be enabled on access switches	l2.dhcp_snooping does not exist on platform 'sonicwall_sonicos': not a SonicOS feature; its equivalent control is MAC-IP anti-spoof
NCSA-EXT-035	Dynamic ARP inspection should be enabled	l2.arp_inspection does not exist on platform 'sonicwall_sonicos': as above -- the equivalent is MAC-IP anti-spoof, mapped at l2.ip_source_guard

NCSA-HTTPS-001	HTTPS management must use TLS 1.2 or higher	HTTPS management is disabled, so its TLS version is moot
NCSA-SSH-001	SSH must be enabled for management access	management.ssh.enabled does not exist on platform 'sonicwall_sonicos': SSH management is bound per zone rather than globally toggled
NCSA-SSH-002	SSH must use protocol version 2	management.ssh.version does not exist on platform 'sonicwall_sonicos': SonicOS speaks SSHv2 only; no version selector exists
NCSA-TEL-001	Telnet must be disabled	management.telnet.enabled does not exist on platform 'sonicwall_sonicos': SonicOS exposes no telnet management service
NCSA-VTY-002	Management access must be restricted by ACL	management.vty.access_class does not exist on platform 'sonicwall_sonicos': management access is zone-bound, not ACL-bound

6. Controls not decided (20)

These controls could not be determined from the configuration supplied. They are neither passes nor failures, and they are counted in the coverage figure so that the limits of this assessment are visible on the front page. They are reported here rather than omitted.

Control	Title	State	Reason
NCSA-AAA-003	Role-based authorisation must be in use	UNKNOWN	authorization.role_based appeared in the config but could not be interpreted
NCSA-CAT-003	Every rule must resolve to a concrete service	UNKNOWN	worst of 329 scoped instance(s); UNKNOWN at Default Access Rule [IPv4#0, Default Access Rule [IPv4#11, Default Access Rule [IPv4#14
NCSA-CAT-007	Defined ACLs must actually be applied	UNKNOWN	firewall.acl_applied was never evaluated by any mapping rule
NCSA-CAT-011	Unused interfaces should be shut down	UNKNOWN	interfaces.shutdown was never evaluated by any mapping rule
NCSA-CAT-012	NetFlow/traffic export should be enabled	UNKNOWN	logging.netflow was never evaluated by any mapping rule
NCSA-CLD-002	No rule may permit all protocols from the internet	UNKNOWN	exposure.any_any_rules appeared in the config but could not be interpreted
NCSA-CLD-003	Unrestricted ingress must be justified and minimal	UNKNOWN	exposure.unrestricted_ingress appeared in the config but could not be interpreted
NCSA-EXT-003	SSH must be bound to a management source interface	UNKNOWN	management.ssh.source_interface was never evaluated by any mapping rule
NCSA-EXT-015	An external authentication method list must be configured	UNKNOWN	authentication.aaa_authentication was never evaluated by any mapping rule
NCSA-EXT-016	Command authorization must be configured	UNKNOWN	authorization.aaa_authorization was never evaluated by any mapping rule
NCSA-EXT-019	Logs must be sourced from a stable interface	UNKNOWN	logging.source_interface was never evaluated by any mapping rule
NCSA-EXT-020	Persistent logging should survive a reboot	UNKNOWN	logging.persistent was never evaluated by any mapping rule
NCSA-EXT-024	SNMPv3 privacy (encryption) must be enabled	UNKNOWN	snmp.v3_priv was never evaluated by any mapping rule
NCSA-EXT-031	Unnecessary services must be disabled	UNKNOWN	services.unused_enabled was never evaluated by any mapping rule
NCSA-EXT-032	ICMP unreachable rate limiting should be enabled	UNKNOWN	management.icmp_ratelimit was never evaluated by any mapping rule

NCSA-EXT-033	Memory thresholds should be configured	UNKNOWN	platform.memory_thresholds was never evaluated by any mapping rule
NCSA-EXT-037	Routing protocol authentication must be enabled	UNKNOWN	routing.ospf_auth was never evaluated by any mapping rule
NCSA-EXT-038	Route filtering should be applied	UNKNOWN	routing.prefix_filtering was never evaluated by any mapping rule
NCSA-NTP-002	NTP must be authenticated	UNKNOWN	time.authenticated was never evaluated by any mapping rule
NCSA-PWD-003	Stored passwords must be encrypted	UNKNOWN	authentication.password_encryption was never evaluated by any mapping rule

Parsing account

Measure	Records
Present in the configuration file	92,635
Read successfully	92,635
Mapped to the compliance schema	8,923
Read but not mapped	83,712
Unreadable	0

Records greatly outnumber settings, because one setting recurs once per rule, per interface or per zone. A record read but not mapped is most often device state with no bearing on a security control — a signature identifier, an object table column, or interface counters.

7. Policy analysis

7.1 Reconstructed policy

The configuration was reconstructed as an object graph: named addresses, services, groups and zones, with every reference followed to the value it resolves to. The findings in section 4 that concern policy are derived from this reconstruction.

Objects reconstructed	846
Policy rules	329
Default action	deny (assumed from platform behaviour, not read from the file)
Evaluation order	first match wins

Object kind	Count
address	312
address group	208
service	255
service group	62
zone	9

7.2 Rule hygiene

Measure	Rules
Examined	329
Fully resolved	150
Not fully resolvable	180

Incomplete analysis. 180 of 329 rules could not be fully resolved, because they reference objects the supplied configuration does not contain. Conclusions about those rules are incomplete, and they are counted here rather than treated as clean.

Observations by kind

Kind	Count
disabled rule	62
disputed shadow	3
orphaned objects	1
overly permissive	55
redundant rule	22
shadowed rule	5

8. Remediation

Review before use. These commands are drawn from vendor documentation for the platform identified in section 1. A configuration file records state, not the commands that produced it, so these sequences cannot be verified against the file assessed. Each step carries a verification command; confirm the effect on the device rather than assuming it. Apply changes through your normal change-control process.

8.1 Suggested commands (12)

NCSA-BAN-001 — A login banner must be displayed

```
configure
administration
login-banner "Authorised access only. Activity is monitored."
commit
```

Verify with: `show administration | grep banner`

NCSA-EXT-029 — Configuration change archiving should be enabled

```
configure
administration
auto-backup enable
commit
```

Verify with: `show administration | grep backup`

NCSA-TIME-001 — Idle session timeout must be 10 minutes or less

```
configure
administration
inactivity-timeout 10
commit
```

Verify with: `show administration | grep inactivity`

NCSA-PWD-001 — Minimum password length must be enforced

```
configure
administration password-policy
minimum-length 15
commit
```

Verify with: `show administration password-policy`

NCSA-EXT-013 — Account lockout must be enabled

```
configure
administration
login-lockout enable
login-lockout-threshold 5
login-lockout-period 5
commit
```

Verify with: `show administration | grep lockout`

NCSA-PLT-002 — Stored credentials and configuration parameters must be encrypted

```
configure
administration
enhanced-password-security enable
commit
```

Verify with: `show administration | grep password-security`

NCSA-PWD-002 — Password complexity must be enforced

```
configure
administration password-policy
complexity alpha-numeric-symbolic
commit
```

Verify with: `show administration password-policy`

NCSA-EXT-022 — Remote syslog must be configured

```
configure
log syslog
add server <SYSLOG-IP> port 514
commit
```

Verify with: `show log syslog server`

NCSA-SNMP-001 — SNMP v1/v2c must not be used

```
configure
administration snmp
version v3-only
commit
```

Verify with: `show administration snmp`

NCSA-EXT-025 — SNMP access must be restricted by ACL

```
configure
administration snmp
management-source-address <MGMT-CIDR>
commit
```

Verify with: show administration snmp

NCSA-EXT-039 — Intrusion prevention should be enabled

```
configure
security-services ips
enable
commit
```

Verify with: show security-services ips

NCSA-EXT-040 — Gateway anti-virus should be enabled

```
configure
security-services gateway-anti-virus
enable
commit
```

Verify with: show security-services gateway-anti-virus

8.3 Findings with no recorded remediation (16)

No command sequence is held for these controls on this platform. They are listed so that this section is not mistaken for a complete remediation plan.

NCSA-CAT-002, NCSA-CLD-001, NCSA-CAT-004, NCSA-EXT-023, NCSA-EXT-001, NCSA-EXT-014, NCSA-SSH-004, NCSA-EXT-006, NCSA-EXT-007, NCSA-EXT-008, NCSA-CAT-006, NCSA-SSH-003, NCSA-EXT-021, NCSA-CAT-010, NCSA-EXT-030, NCSA-EXT-042

8.4 Consolidated sequence

```
! NCSA remediation -- review before running.
! platform: sonicwall_sonicos
! [MEDIUM] NCSA-BAN-001: A login banner must be displayed
configure
administration
login-banner "Authorised access only. Activity is monitored."
commit
! verify: show administration | grep banner
!
! [MEDIUM] NCSA-EXT-029: Configuration change archiving should be enabled
configure
administration
auto-backup enable
commit
! verify: show administration | grep backup
!
! [HIGH] NCSA-TIME-001: Idle session timeout must be 10 minutes or less
configure
administration
inactivity-timeout 10
commit
! verify: show administration | grep inactivity
!
! [MEDIUM] NCSA-PWD-001: Minimum password length must be enforced
configure
administration password-policy
minimum-length 15
commit
! verify: show administration password-policy
!
! [MEDIUM] NCSA-EXT-013: Account lockout must be enabled
configure
administration
login-lockout enable
login-lockout-threshold 5
login-lockout-period 5
commit
! verify: show administration | grep lockout
!
! [MEDIUM] NCSA-PLT-002: Stored credentials and configuration parameters must be encrypted
configure
administration
enhanced-password-security enable
commit
! verify: show administration | grep password-security
!
! [MEDIUM] NCSA-PWD-002: Password complexity must be enforced
configure
administration password-policy
complexity alpha-numeric-symbolic
commit
! verify: show administration password-policy
!
! [MEDIUM] NCSA-EXT-022: Remote syslog must be configured
configure
log syslog
```

```
add server <SYSLOG-IP> port 514
commit
! verify: show log syslog server
!
! [HIGH] NCSA-SNMP-001: SNMP v1/v2c must not be used
configure
administration snmp
version v3-only
commit
! verify: show administration snmp
!
! [MEDIUM] NCSA-EXT-025: SNMP access must be restricted by ACL
configure
administration snmp
management-source-address <MGMT-CIDR>
commit
! verify: show administration snmp
!
! [MEDIUM] NCSA-EXT-039: Intrusion prevention should be enabled
configure
security-services ips
enable
commit
! verify: show security-services ips
!
! [LOW] NCSA-EXT-040: Gateway anti-virus should be enabled
configure
security-services gateway-anti-virus
enable
commit
! verify: show security-services gateway-anti-virus
!
```

9. Method and limitations

9.1 How this assessment was produced

The configuration file was identified by content, parsed with the mapping pack for its platform, and normalised into a vendor-neutral model of security settings. Controls were then evaluated against that model. No vendor-specific logic exists in the control layer, so the same control is assessed identically across platforms.

9.2 How the score is calculated

The compliance score is the weighted proportion of passing controls among those that could be decided. Controls that are not applicable, not decided, require manual review, or errored are excluded from the calculation and are reported separately in sections 5 and 6. Partially met controls count as half. Coverage states what proportion of the catalogue was decided, and the score is meaningful only when read with it.

9.3 What this assessment does not cover

- It examines a configuration file only. Running state, live traffic, installed firmware integrity and physical security are outside its scope.
- A control reported as not decided is a limit of this assessment. It must not be recorded as a pass.
- Where policy rules reference objects absent from the supplied file, any conclusion about those rules is incomplete. The count of such rules is given in section 7.
- Remediation commands are suggestions derived from vendor documentation and are not verified against this device.

9.4 Framework references

Findings cite control identifiers from NIST SP 800-53, DISA STIG, CIS Benchmarks and ISO/IEC 27001 where a mapping exists. CIS and ISO material is copyrighted; it is cited by identifier and short title only, and no text from either is reproduced in this report.